EnzymeTuner: Comprehensive User Tutorial

A Standard Operating Procedure for Rational Thermostabilization and Kinetic Parameter Engineering (K_m & V_{max})

1. Executive Summary & Biophysical Principles

Simultaneously engineering an enzyme for elevated thermostability (higher melting temperature T_m and extended operational half-life) and optimized catalytic performance (lower K_m, higher k_{cat} or V_{max}) is complicated by the fundamental **activity-stability trade-off**[cite: 8]. Active site residues and catalytic loops require conformational mobility to bind substrates, lower activation barriers ($\Delta G^{\ddagger}$), and release products, whereas global thermostabilization generally involves structural rigidification[cite: 8].

EnzymeTuner's Spatial Partitioning Solution: To decouple these competing forces, the protein coordinates are partitioned into three concentric geometric zones relative to the catalytic center[cite: 8]:

- **Catalytic Core (0 – 4.5 Å):** Strictly conserved residues directly coordinating chemical mechanisms[cite: 8]. Locked to avoid catalytic inactivation[cite: 8].
- **Kinetic Tuning Shell (4.5 – 8.5 Å):** Second-shell residues lining the binding cleft and gating loops[cite: 8]. Targeted to tune shape complementarity and transition-state electrostatics without destroying chemistry[cite: 8].
- **Scaffold Zone (> 8.5 Å):** Outer surface loops and core hydrophobic packings[cite: 8]. Screened to optimize global thermodynamic folding free energy ($\Delta\Delta G_{folding} < 0$) with minimal active-site perturbation[cite: 8].

2. Pipeline Architecture & Multi-Engine Integration

EnzymeTuner integrates evolutionary learning, physical force fields, molecular docking, and spatial geometry[cite: 8]:

Pipeline Stage	Software / Engine	Physical / Biological Metric
1. Evolutionary Screen	Meta AI ESM-2 (35M Transformer)	Zero-shot log-likelihood ratio (LLR ≥ -1.0) eliminates aggregation/folding defects[cite: 8].
2. Scaffold Stability	FoldX 5.0 (RepairPDB + BuildModel)	$\Delta\Delta G_{folding}$ (kcal/mol)[cite: 8]. Identifies stabilizing substitutions ($\Delta\Delta G < -1.0$)[cite: 8].

Pipeline Stage	Software / Engine	Physical / Biological Metric
3. Kinetic Substrate Fit	AutoDock Vina + OpenBabel	$\Delta\Delta G_{bind}$ vs. Wild-Type (kcal/mol)[cite: 8]. Calibrates substrate affinity & TS fit[cite: 8].
4. Combinatorial Design	Spatial Distance Matrix & Additivity	Pairs non-conflicting scaffold & pocket mutations (≥ 9 Å Cα separation)[cite: 8].

3. Environment Setup & Software Prerequisites

Install the required numerical, machine-learning, and structural biology tools into an isolated Conda environment[cite: 8]:

```
# 1. Create and activate dedicated environment
conda create -n enzymetuner python=3.10 -y
conda activate enzymetuner

# 2. Install numerical, structural, and docking engines
conda install -c conda-forge openbabel vina biopython pandas matplotlib -y
pip install torch fair-esm

# 3. Install FoldX (Academic binary available at foldxsuite.crg.eu)
# Copy 'foldx' to your PATH (e.g. /usr/local/bin/foldx) or provide path via --fol
```

4. Input File Preparation Guidelines

- **A. Enzyme Structure (PDB):** Clean the coordinate file prior to execution[cite: 8]. Strip crystallization waters, salts, and non-covalent additives (e.g., glycerol, PEG, sulfate)[cite: 8]. Retain critical structural metal ions (e.g., Zn^{2+}, Mg^{2+}, $Fe^{2+/3+}$)[cite: 8].
- **B. Active Site Identification:** Query the Mechanism and Catalytic Site Atlas (M-CSA) or UniProt for residue IDs[cite: 8]. Numbering passed via `--active_site` must strictly match the chain residue numbering in the PDB file[cite: 8].
- **C. Ligand / Transition-State Model (SDF/MOL2):** Supply a 3D structural model of the substrate or transition-state analog[cite: 8]. The script automatically applies Gasteiger partial charges and standard physiological protonation at pH 7.4 via OpenBabel[cite: 8].

5. Execution Syntax & Operational Recipes

Execute the end-to-end design run from your terminal shell[cite: 9]:

```
python enzymetuner.py \
  --pdb 1XYZ_clean.pdb \
```

```
    --chain A \
    --active_site 35 52 108 \
    --ligand substrate.sdf \
    --foldx_bin /usr/local/bin/foldx \
    --max_foldx 40 \
    --max_vina 20 \
    --out single_candidates.csv \
    --out_combos combinatorial_designs.csv \
    --plot optimization_landscape.png
```

6. Interpreting Output Tables & Selection Rules

The pipeline generates two structured CSV files and one publication-grade scatter plot[cite: 9]:

- `single_candidates.csv` : Evaluates single-point variants across `Target_Class` , `FoldX_ddG_kcal_mol` , and `Delta_Vina_Affinity_kcal_mol` [cite: 9].
- `combinatorial_designs.csv` : Proposes multi-point designs combining verified scaffold stabilizers with kinetic pocket tuners with guaranteed spatial independence[cite: 9].
- `optimization_landscape.png` : 4-quadrant plot mapping folding stability (x-axis) vs. affinity change (y-axis) colored by ESM-2 fitness[cite: 9].

> **Experimental Hit Selection Strategy:**
>
> • **Synergistic Single Hits:** Focus on points in the lower-left quadrant ($\Delta\Delta G_{folding} < 0$ and $\Delta\Delta G_{bind} < -0.4$ kcal/mol)[cite: 9].
>
> • **Modular Combinations (Recommended):** Select 2 top scaffold stabilizers ($\Delta\Delta G < -1.2$ kcal/mol) and 1 pocket tuner ($\Delta\Delta G_{bind} < -0.7$ kcal/mol)[cite: 9]. The combined construct establishes a substantial thermal buffer (+5°C to +12°C T_m) while driving faster turnover[cite: 9].

7. Wet-Lab Experimental Validation Protocol

Follow this standardized screening cascade to validate candidate designs experimentally[cite: 9]:

1. **Expression & Soluble Yield:** Screen expression in *E. coli* BL21(DE3)[cite: 9]. SDS-PAGE of soluble vs. pellet fractions verifies that mutations do not compromise folding[cite: 9].
2. **Thermostability Assay (DSF / CD):** Quantify melting temperature (T_m) shifts using Differential Scanning Fluorimetry (DSF, SYPRO Orange) or Far-UV Circular Dichroism (CD) at 222 nm[cite: 9].
3. **Steady-State Enzyme Kinetics:** Determine initial rates across 8–12 substrate concentrations[cite: 9]. Fit data to the Michaelis-Menten model to extract K_m, V_{max}, and k_{cat}/K_m[cite: 9].

4. **Thermal Inactivation Profiles:** Incubate WT and variants at elevated temperatures (e.g., 50°C, 60°C) over time (0, 15, 30, 60 min) to determine residual activity half-life ($t_{1/2}$)[cite: 9].

8. Troubleshooting & Common Pitfalls

- **FoldX Error during RepairPDB:** Remove non-standard modified amino acids, water molecules, or heteroatoms before initiating the run[cite: 9].
- **Docking Centroid Off-Center:** Ensure the residue numbers passed to `--active_site` match the PDB chain sequentially[cite: 9].
- **Loss of Catalytic Activity:** Never include residues within 4.5 Å of the catalytic triad/dyad in mutagenesis libraries[cite: 9].